

QR-UOV without Rejection Sampling: Security Analysis and High-Speed Implementation

Hiroshi Amagasa^{*1}, Hiroki Furue², Rei Ueno^{3,1}, and Naofumi Homma¹

¹Tohoku University, Japan

²NTT Social Informatics Laboratories, Japan

³Kyoto University, Japan

Abstract

QR-UOV is a multivariate signature scheme derived from UOV that achieves compact public keys by exploiting quotient-ring structure, making it a promising candidate for post-quantum digital signatures. In QR-UOV, most parts of the public map are constructed by extending the public key seed using PRG. This public key expansion for QR-UOV includes rejection sampling to generate coefficients uniformly over $\mathbb{F}_q$, since QR-UOV uses a small odd-prime base field. However, this rejection sampling introduces extra data movement and irregular control flow. For the recommended parameter set, public-key expansion accounts for nearly 90% of the QR-UOV verification time.

In this paper, we propose No Rejection Sampling (NoRS) QR-UOV, which removes rejection sampling from public-key expansion and leaves the generation of secret-dependent coefficients unchanged. Concretely, the rejected value q is deterministically mapped to 0, which simplifies coefficient generation but introduces a slight bias in the resulting coefficient distribution. We evaluate the security impact of this modification through both theoretical and concrete analyses. Our results indicate that, for the proposed parameter sets, NoRS QR-UOV preserves the claimed security levels.

On the implementation side, we develop a high-speed implementation of NoRS QR-UOV for x86 processors with AES-NI and AVX2. Benchmark results on a Skylake platform show that NoRS consistently accelerates QR-UOV at all security levels, with the largest gain in signature verification. For the AES-128-based implementation, the verification cost is reduced from 0.43 to 0.30 Mcycles at security level I, with similar improvements at levels III and V, corresponding to about 1.4 $\times$ speedup. Overall, the results suggest that relaxing coefficient uniformity in public-key expansion is a practical and effective design choice for QR-UOV.

Keywords: post-quantum cryptography, multivariate public-key cryptography, QR-UOV, rejection sampling, AES-NI

1 Introduction

Practical deployment of post-quantum cryptography calls for not only security evaluation but also implementation efficiency. This requirement is particularly important for digital signatures, where verification throughput can be a decisive factor in real-world applications. This viewpoint is also reflected in the ongoing NIST standardization process for additional digital signatures, which aims to diversify the standardized portfolio beyond structured-lattice schemes and has explicitly highlighted interest in schemes with short signatures and fast verification.

QR-UOV [1] is a multivariate signature scheme derived from UOV [2]. It is one of the candidates selected for Round 2 of the NIST standardization process for additional digital signatures. The main feature of QR-UOV is its quotient-ring structure, which mitigates a major drawback of UOV, namely

^{*}hiroshi.amagasa.q7@dc.tohoku.ac.jp

its large public key. By exploiting the quotient-ring structure, QR-UOV reduces the public-key size by about 50% compared with UOV, while maintaining security. NIST’s first-round status report [3] notes that, although QR-UOV is slower than UOV, its performance is still competitive. The report also states that its quotient-ring structure requires further study, and that its implementation leaves room for future optimization.

A previous implementation study [4] improved QR-UOV performance, and also showed that QR-UOV verification is dominated by public-key expansion. For a recommended parameter set $(q, \ell) = (127, 3)$, nearly 90% of the total verification time comes from public-key expansion, with AES-CTR and rejection sampling accounting for 61% and 27%, respectively. Hence, coefficient generation during expansion is a major implementation bottleneck of QR-UOV. This bottleneck closely relates to a distinctive algebraic feature of QR-UOV. Unlike other multivariate candidates in the NIST standardization process for additional digital signatures—UOV [5], SNOVA [6], and MAYO [7]—all of which use characteristic-two fields, QR-UOV uses a small odd-prime base field. As a result, uniform coefficient generation over $\mathbb{F}_q$ in public-key expansion requires rejection sampling. This scheme-specific step introduces extra data movement and irregular control flow, and is a major source of software overhead in verification latency.

In this work, to address this bottleneck, we propose No Rejection Sampling (NoRS) QR-UOV, a variant that removes rejection sampling from public-key expansion while leaving secret-dependent coefficient generation unchanged. Concretely, the rejected value in standard QR-UOV is deterministically mapped to 0. This modification simplifies and regularizes coefficient generation, but it introduces a slight bias in the coefficient distribution. Hence, NoRS QR-UOV must be evaluated from both security and implementation viewpoints. This paper makes three main contributions spanning security analysis, implementation, and performance evaluation.

- We analyze the coefficient bias introduced by NoRS and evaluate its security impact through both theoretical and concrete security analyses. In particular, we show that the proposed parameter sets preserve the claimed security levels from the original QR-UOV.
- We develop a high-speed implementation of NoRS QR-UOV on x86 processors with AES-NI and AVX2, including a fused expansion-and-dot-product routine for verification.
- We demonstrate that NoRS consistently accelerates QR-UOV at all security levels, with the largest gain in signature verification.

2 Preliminaries

2.1 UOV and QR-UOV

Multivariate public-key cryptography is based on the hardness of solving systems of multivariate quadratic equations over a finite field. In UOV-type signature schemes, the private key consists of a structured central quadratic map $\mathcal{F} = (f_1, \dots, f_m) : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ and a random invertible linear map $\mathcal{S} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^n$ while the public key is their composition $\mathcal{P} = \mathcal{F} \circ \mathcal{S}$. A signature $s \in \mathbb{F}_q^n$ on a message M satisfies $\mathcal{P}(s) = t$, where $t \in \mathbb{F}_q^m$ is the vector derived from the message digest and salt. Verification checks this equality.

In UOV [2], the variables are divided into v vinegar variables and m oil variables, with $n = v + m$. Each quadratic polynomial f_i can be represented by an $n \times n$ matrix F_i of the form

$$F_i = \begin{pmatrix} *_{v \times v} & *_{v \times m} \\ *_{m \times v} & 0_{m \times m} \end{pmatrix},$$

where the zero lower-right submatrix means that no products of two oil variables appear in f_i . Hence, once the vinegar variables are fixed, the resulting equations are linear in the oil variables. Let S denote the matrix representation of $\mathcal{S}$. Then the corresponding public-key matrices are given by $P_i = S^\top F_i S$.

QR-UOV [1] is a structured variant of UOV that reduces the public-key size by introducing quotient-ring structure. Let $\mathbb{F}_{q^\ell} = \mathbb{F}_q[x]/(f)$ for an irreducible polynomial $f \in \mathbb{F}_q[x]$ of degree ℓ . For each $g \in \mathbb{F}_{q^\ell}$, define an $\ell \times \ell$ matrix $\Phi_g^f \in \mathbb{F}_q^{\ell \times \ell}$ by

$$(1, x, \dots, x^{\ell-1})\Phi_g^f = (g, xg, \dots, x^{\ell-1}g) \pmod{f}.$$

This gives an injective ring homomorphism from $\mathbb{F}_{q^\ell}$ into $\mathbb{F}_q^{\ell \times \ell}$, so that each structured $\ell \times \ell$ block is determined by only ℓ coefficients over $\mathbb{F}_q$, rather than by ℓ^2 independent matrix entries. In QR-UOV, the public key matrices P_i with $i \in [m]$ are constructed as block matrices whose every component is given as $W\Phi_g^f$ with $g \in \mathbb{F}_{q^\ell}$, where $W \in \mathbb{F}_q^{\ell \times \ell}$ satisfies the condition that $W\Phi_g^f$ is symmetric for any $g \in \mathbb{F}_{q^\ell}$. As a result, QR-UOV achieves a more compact public-key representation than UOV. Under the vinegar-oil partition, each public-key matrix $P_i \in \mathbb{F}_q^{n \times n}$ is partitioned as

$$P_i = \begin{pmatrix} P_{i,1} & P_{i,2} \\ P_{i,2}^\top & P_{i,3} \end{pmatrix},$$

where $P_{i,1} \in \mathbb{F}_q^{v \times v}$, $P_{i,2} \in \mathbb{F}_q^{v \times m}$, $P_{i,3} \in \mathbb{F}_q^{m \times m}$. Since P_i is symmetric, both $P_{i,1}$ and $P_{i,3}$ are also symmetric. The compressed public key consists of the matrices $P_{i,3}$ and a public seed. The remaining matrices $P_{i,1}$ and $P_{i,2}$ are expanded from the seed by $\text{Expand}_{\text{pk}}$, which uses a PRG instantiated by SHAKE or AES-CTR.

2.2 QR-UOV verification procedure

To prepare for the verification optimizations described later, we briefly review the QR-UOV verification procedure and fix the notation used throughout the paper. We denote by $\mathbf{p}_{i,1}$, $\mathbf{p}_{i,2}$, and $\mathbf{p}_{i,3}$ the compact coefficient-vector representations of the structured matrices $P_{i,1}$, $P_{i,2}$, and $P_{i,3}$, respectively.

For a signature vector $\mathbf{s}$, the public map evaluation $\mathcal{P}(\mathbf{s}) = \mathbf{t}$ can be written in a component-wise manner as

$$t_i = \mathbf{s}^\top P_i \mathbf{s} \quad (i \in [m]).$$

Following the previous QR-UOV implementation study [4], each coordinate evaluation can be expressed as a sum of dot products between vectors precomputed from the signatures and vectors regenerated from the compressed public key. Specifically, after computing $(\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3)$ from $\mathbf{s}$, the public-key expansion expands $(\mathbf{p}_{i,1}, \mathbf{p}_{i,2})$ from seed_{pk} and evaluates

$$t'_i = \mathbf{u}_1 \cdot \mathbf{p}_{i,1} + \mathbf{u}_2 \cdot \mathbf{p}_{i,2} + \mathbf{u}_3 \cdot \mathbf{p}_{i,3},$$

which is equivalent to computing $\mathbf{s}^\top P_i \mathbf{s}$. Hence, QR-UOV verification can be viewed as public-key expansion followed by dot-product evaluation. Algorithm 1 shows how this view is realized in the verification procedure.

2.3 Rejection sampling in public-key expansion

In QR-UOV, the base-field parameter q is chosen from $\{7, 31, 127\}$, so $q = 2^k - 1$ for $k \in \{3, 5, 7\}$. This Mersenne-prime form is convenient for arithmetic over $\mathbb{F}_q$, but it also affects coefficient generation in public-key expansion. To generate coefficients uniformly over $\mathbb{F}_q$ from pseudorandom bits, the public-key expansion first applies a bitmask to obtain k -bit values in $\{0, \dots, 2^k - 1\} = \{0, \dots, q\}$. Since the valid coefficient range is $\{0, \dots, q - 1\}$, the unique out-of-range value q must be rejected. Thus, standard QR-UOV uses rejection sampling during public-key expansion to obtain a uniform distribution over $\mathbb{F}_q$.

In the Round-2 QR-UOV specification [8], rejection sampling is realized by overwriting each occurrence of q in the prefix with a non- q value from the remaining sequence. Although the rejection probability is only $1/2^k$, this replacement step still introduces irregular control flow and makes SIMD parallelization less straightforward.

Algorithm 1 Verify(M, pk, σ)

Input: message $M \in \mathbb{B}^*$, public key $\text{pk} \in \{0, 1\}^\lambda \times \left(\mathbb{F}_q^{M \times M}\right)^m$, and signature $\sigma \in \{0, 1\}^\lambda \times \mathbb{F}_q^m$
Output: accept or reject

- 1: $(\text{seed}_{\text{pk}}, \{p_{i,3}\}_{i \in [m]}) \leftarrow \text{pk}$
- 2: $(r, s) \leftarrow \sigma$
- 3: $(u_1, u_2, u_3) \leftarrow \text{ComputeCoefficients}(s)$
- 4: $\mu \leftarrow \text{SHAKE256}(\text{seed}_{\text{pk}} \parallel \text{BytesToBits}(M), 512)$
- 5: $t \leftarrow \text{Hash}(\mu, r)$ $\triangleright t, t' \in \mathbb{F}_q^m$
- 6: **for** i from 1 to m **do**
- 7: $(p_{i,1}, p_{i,2}) \leftarrow \text{Expand}_{\text{pk}}(\text{seed}_{\text{pk}}, i)$
- 8: $t'_i \leftarrow u_1 \cdot p_{i,1} + u_2 \cdot p_{i,2} + u_3 \cdot p_{i,3}$ $\triangleright t'_i \leftarrow s^\top P_i s$
- 9: **end for**
- 10: **return** accept if $t = t'$ and reject otherwise.

3 Proposed scheme: NoRS QR-UOV

Our proposed scheme follows the QR-UOV Round-2 specification [8] and modifies only the rejection-sampling step in public-key expansion from the public seed seed_{pk} . All secret-dependent coefficient generation is left unchanged.

Let $q = 2^k - 1$, and let $r \in \{0, \dots, q\}$ be the k -bit value generated from the PRG output during public-key expansion. Instead of rejecting the case $r = q$, NoRS maps the output coefficient $a \in \mathbb{F}_q$ by

$$a := r \bmod q.$$

Equivalently,

$$a = \begin{cases} 0 & \text{if } r = q, \\ r & \text{otherwise.} \end{cases}$$

At the implementation level, the exceptional value $r = q$ can be kept unchanged as a non-canonical representative of $0 \in \mathbb{F}_q$. This modification eliminates the rejection step and induces a slight bias in the distribution of the generated coefficients. The resulting distribution is

$$\Pr[a = 0] = \frac{2}{2^k}, \quad \Pr[a = x] = \frac{1}{2^k} \quad (x \in \mathbb{F}_q, x \neq 0).$$

For the recommended parameter set $q = 127$ (i.e., $k = 7$), the only deviation from uniformity is that 0 occurs twice among the 128 possible 7-bit candidates.

In the remainder of this paper, we use the AES-CTR instantiation of $\text{Expand}_{\text{pk}}$ and the recommended Round-2 QR-UOV instances. For each security level, we consider the recommended instance with $q = 127$ and $\ell = 3$ listed below.

- $(q, v, m, \ell) = (127, 156, 54, 3)$ for security level I,
- $(q, v, m, \ell) = (127, 228, 78, 3)$ for security level III,
- $(q, v, m, \ell) = (127, 306, 105, 3)$ for security level V.

4 Security evaluation

This section analyzes the security of the proposed NoRS QR-UOV presented in Section 3. We first provide a security proof for NoRS QR-UOV and then discuss the hardness of the underlying problems.

4.1 Security proof

This subsection provides a proof for the EUF-CMA security of the proposed NoRS QR-UOV. Recall that, in the specification [8], the security of QR-UOV is reduced to the following two problems: distinguishing a UOV public map from a random MQ map, and solving the MQ problem with the QR structure. For the security proof of the proposed scheme, we introduce variants of the underlying problems that incorporate NoRS as follows:

- **NoRS UOV problem.** For parameters (q, v, o, m) , consider the UOV public map $\mathcal{P}$ with v vinegar variables, o oil variables, and m equations over $\mathbb{F}_q$, where each coefficient corresponding to $P_{i,1}$ and $P_{i,2}$ for $i \in [m]$ is obtained by NoRS. Let $\mathcal{P}'$ be an MQ map with $v + o$ variables and m equations over $\mathbb{F}_q$ where each coefficient is obtained by NoRS. Then, the NoRS UOV problem asks to distinguish these two MQ maps $\mathcal{P}$ and $\mathcal{P}'$.
- **NoRS QR-MQ problem.** For parameters (q, n, m, ℓ) , consider the MQ map $\mathcal{P} : \mathbb{F}_q^n \rightarrow \mathbb{F}_q^m$ defined by

$$\mathcal{P}(\mathbf{x}) = (\mathbf{x}^\top P_1 \mathbf{x}, \dots, \mathbf{x}^\top P_m \mathbf{x})^\top,$$

where $P_1, \dots, P_m$ are given as block matrices whose every component has the form of $W\Phi_g^f$ and each coefficient of $g \in \mathbb{F}_{q^\ell}$ is obtained by NoRS. Then, the NoRS QR-MQ problem asks to solve $\mathcal{P}(\mathbf{x}) = \mathbf{t}$ for a randomly chosen $\mathbf{t} \in \mathbb{F}_q^m$.

Then, we can reduce the EUF-CMA security of the NoRS QR-UOV to the above two problems as follows:

Theorem 1 (EUF-CMA security of NoRS QR-UOV). *For any quantum EUF-CMA adversary $\mathcal{A}_{\text{cma}}$ against the NoRS QR-UOV with parameters (q, v, m, ℓ) issuing at most q_{sign} classical queries to the signing oracle and q_{qro} (quantum) random oracle queries, there exist adversaries $\mathcal{B}_{\text{uov}}$ and $\mathcal{B}_{\text{mq}}$ against the NoRS UOV problem with parameters $(q^\ell, v/\ell, m/\ell, m)$ and the NoRS QR-MQ problem with parameters $(q, (v + m), m, \ell)$, respectively, such that*

$$\begin{aligned} \text{Adv}(\mathcal{A}_{\text{cma}}) &\leq (2q_{\text{qro}} + 1)^2 (\text{Adv}(\mathcal{B}_{\text{uov}}) + \text{Adv}(\mathcal{B}_{\text{mq}})) \\ &+ \frac{3}{2} q'_{\text{sign}} \sqrt{\frac{q'_{\text{sign}} + q_{\text{qro}} + 1}{2^\lambda}} \\ &+ 2(q_{\text{qro}} + 2) \sqrt{\frac{q'_{\text{sign}} - q_{\text{sign}}}{2^\lambda}} + \frac{q'_{\text{sign}} + m + 1}{2^\lambda}, \end{aligned}$$

where Adv denotes the advantage of adversaries against each problem and q'_{sign} is a bound on the total number of random oracle queries in all the signing queries. The running times of $\mathcal{B}_{\text{uov}}$ and $\mathcal{B}_{\text{mq}}$ are essentially the same as that of $\mathcal{A}_{\text{cma}}$.

Proof. The proof from the non-invertibility to the EUF-CMA security directly follows [8, Lemma 3]. The reduction from the hardness of the NoRS UOV and QR-MQ problems to the non-invertibility is proven by simply replacing the two problems by the NoRS variants of them in [8, Lemma 5]. $\square$

4.2 Concrete security evaluation

In this study, we adopt the same main parameter sets as in the original QR-UOV for the proposed NoRS QR-UOV, corresponding to security levels I, III, and V. This subsection shows that these parameters also achieve the claimed security levels for NoRS QR-UOV.

In the specification [8], the parameters are chosen such that the complexity of all known attacks on QR-UOV exceeds the criteria for the claimed security levels, as shown in Table 1. In the following, we ensure the security of NoRS QR-UOV with these parameters by comprehensively analyzing the

Table 1: The complexity of the existing attacks on the main parameters of QR-UOV in Table 14 of [8]

SL	(q, v, m, ℓ)	complexity					
		Claw	Dir.	Rec.	RM	KS	Int.
I	(127,156,54,3)	201	150	164	158	718	460
III	(127,228,78,3)	286	211	231	219	1056	653
V	(127,306,105,3)	380	279	291	277	1414	851

complexity of these attacks against NoRS QR-UOV. In particular, we discuss the effect of NoRS on the efficiency of the claw-finding attack, the direct attack on the QR-MQ problem, and two key-recovery attacks on the UOV problem, the reconciliation [9] and rectangular MinRank [10, 11] attacks. We do not discuss the Kipnis-Shamir [12] and intersection [10] attacks, since their complexities are significantly higher than those of the other attacks as shown in Table 1, and thus they do not affect the security of the parameters even when NoRS is applied.

Claw-finding attack For a given message representative μ , the claw-finding attack computes $\mathcal{P}(\mathbf{x})$ for multiple $\mathbf{x} \in \mathbb{F}_q^n$ and $\text{Hash}(\mu, r)$ for multiple bit strings r . The attack then forges a signature by finding a collision between these two sets. The complexity of the attack mainly depends on the probability that $\mathcal{P}(\mathbf{x}) = \text{Hash}(\mu, r)$ holds for randomly chosen $\mathbf{x}$ and r . In NoRS QR-UOV, the distribution of $\mathcal{P}$ differs from that in the original scheme, whereas $\text{Hash}(\mu, r)$ outputs a uniformly random element of $\mathbb{F}_q^m$. Therefore, the probability that $\mathcal{P}(\mathbf{x}) = \text{Hash}(\mu, r)$ holds remains the same as in the original QR-UOV, namely q^{-m} . Consequently, the complexity of the claw-finding attack on NoRS QR-UOV is the same as that on the original QR-UOV.

Direct attack We evaluate the complexity of the direct attack on the NoRS QR-MQ problem, which simply solves the MQ system algebraically. In general, the complexity of solving an MQ system with n variables and m equations is given by

$$3 \binom{n+2}{2} \binom{n+D}{D}^2,$$

where D denotes the solving degree determined for the given system. Let I be the ideal generated by the given quadratic polynomials, and let $I_{\leq d}$ denote the subspace of I consisting of polynomials of degree at most d . Then, the solving degree D is defined as the minimum degree d such that the codimension of $I_{\leq d}$ in $\mathbb{F}_q[\mathbf{x}]_{\leq d}$ is equal to 1, assuming that the solution is unique.

To compare the complexity of the direct attack against the original and NoRS QR-MQ problems, we compare the solving degrees of the two systems experimentally. Table 2 shows that the codimension of $I_{\leq d}$ for $2 \leq d \leq D$ in the NoRS QR-MQ problem coincides with that of the original QR-MQ problem for the considered parameters. These results indicate that the complexity of the direct attack on the NoRS QR-MQ problem is the same as that on the original QR-MQ problem.

Reconciliation attack The reconciliation attack on the underlying UOV problem with parameters $(q^\ell, v/\ell, m/\ell, m)$ solves the system of

$$\mathbf{x}^\top \bar{P}_i \mathbf{x} = 0 \quad (i \in [m]),$$

where $\bar{P}_i \in \mathbb{F}_{q^\ell}^{n/\ell \times n/\ell}$ denotes the public key matrix of UOV over the extension field. The dimension of the solution space is m/ℓ . Hence, the system can be regarded as one with $v/\ell + 1$ variables and m homogeneous quadratic equations over $\mathbb{F}_{q^\ell}$.

The complexity of the reconciliation attack depends on the solving degree D as in the direct attack. We therefore compare the solving degree of the attack for the original and NoRS UOV problems by

Table 2: The comparison of the codimension of $I_{\leq d}$ in $\mathbb{F}_q[\mathbf{x}]_{\leq d}$ for the direct attack against the original and NoRS QR-MQ problems with $q = 127$ and $\ell = 3$

(n, m)	QR-MQ	d						
		2	3	4	5	6	7	8
(12,14)	original	77	273	637	1001	1001	429	1
	NoRS	77	273	637	1001	1001	429	1
(15,18)	original	118	528	1581	3264	4488	3264	1
	NoRS	118	528	1581	3264	4488	3264	1
(15,19)	original	117	512	1463	2736	2907	1	
	NoRS	117	512	1463	2736	2907	1	
(15,20)	original	116	496	1346	2224	1444	1	
	NoRS	116	496	1346	2224	1444	1	

Table 3: The comparison of the codimension of I_d in $\mathbb{F}_{q^\ell}[\mathbf{x}]_d$ for the reconciliation attack against the original and NoRS UOV problems with parameters $(q^\ell, v/\ell, m/\ell, m)$ where $q = 127$ and $\ell = 3$

(v, m)	UOV	d					
		2	3	4	5	6	7
(30,12)	original	54	154	275	297	132	1
	NoRS	54	154	275	297	132	1
(33,15)	original	63	184	300	168	1	
	NoRS	63	184	300	168	1	
(36,18)	original	73	221	335	1		
	NoRS	73	221	335	1		
(36,21)	original	70	182	119	1		
	NoRS	70	182	119	1		

experimentally evaluating the codimension of the ideal generated by the equations, as shown in Table 3. Unlike in the direct attack case, we consider the subspace $I_d \subseteq I$ consisting of degree- d polynomials, since the system to be solved in the key recovery attacks is homogeneous. Table 3 shows that the solving degree takes the same value for the reconciliation attacks on the original and NoRS UOV problems with parameters $(q^\ell, v/\ell, m/\ell, m)$. This indicates that the complexity of the reconciliation attack on the NoRS UOV problem remains the same as that on the original UOV problem.

Rectangular MinRank attack We analyze the complexity of the rectangular MinRank attack on the underlying NoRS UOV problem with parameters $(q^\ell, v/\ell, m/\ell, m)$. The attack consists of the following two steps:

1. Transform the public key matrices over $\mathbb{F}_{q^\ell}$ into a set of n/ℓ matrices in $\mathbb{F}_{q^\ell}^{n/\ell \times m}$.
2. Find a linear combination of the n/ℓ matrices having rank v/ℓ by solving a MinRank problem.

From the viewpoint of complexity, the second step is dominant, and the support minors method [13] is known to be one of the most efficient approaches to solving the MinRank problem. The method introduces $v/\ell + 1$ linear variables $\mathbf{x}$ corresponding to elements of the secret key S and $\binom{m}{v/\ell}$ support variables $\mathbf{c}$. For these variables, we solve a system consisting of $\binom{m}{v/\ell+1} n/\ell$ homogeneous bilinear equations in $\mathbf{x}$ and $\mathbf{c}$, and m homogeneous quadratic equations in $\mathbf{x}$. The complexity of the support minors approach depends

Table 4: The comparison of the codimension of $I_{d,1}$ in $\mathbb{F}_{q^\ell}[\mathbf{x}, \mathbf{c}]_{d,1}$ for the rectangular MinRank attack against the original and NoRS UOV problems with parameters $(q^\ell, v/\ell, m/\ell, m)$ where $q = 127$ and $\ell = 3$

(v, m)	UOV	d			
		2	3	4	5
(15,6)	original	49	21	1	
	NoRS	49	21	1	
(18,9)	original	1			
	NoRS	1			
(21,9)	original	316	1		
	NoRS	316	1		
(24,9)	original	226	369	246	1
	NoRS	226	369	246	1

on the solving degree of the resulting system, as in the direct and reconciliation attacks. In practice, the system is typically solved by adjusting only the degree of $\mathbf{x}$ while fixing the degree of $\mathbf{c}$ to 1 for efficiency.

We then estimate the complexity of the rectangular MinRank attack on the NoRS UOV problem by comparing the solving degree D . Table 4 shows the codimension of $I_{d,1}$ in $\mathbb{F}_{q^\ell}[\mathbf{x}, \mathbf{c}]_{d,1}$, where $I_{d,1}$ denotes the set of polynomials in the ideal I generated by the equations to be solved whose degree in $\mathbf{x}$ is d and whose degree in $\mathbf{c}$ is 1. The table shows that the solving degree takes the same value for the rectangular MinRank attacks on the original and NoRS UOV problems with parameters $(q^\ell, v/\ell, m/\ell, m)$. This indicates that the complexity of the rectangular MinRank attack on the NoRS UOV problem remains the same as that on the original UOV problem.

We studied the impact of the coefficient bias introduced by NoRS on the security of QR-UOV with respect to existing attacks. From these analyses, we conclude that the proposed parameter sets of NoRS QR-UOV achieve the claimed security levels.

5 Efficient implementation of NoRS QR-UOV verification

As described in Section 2.2, QR-UOV verification consists of public-key expansion followed by dot-product evaluation. In Algorithm 1, for each coordinate i , the verifier regenerates $(\mathbf{p}_{i,1}, \mathbf{p}_{i,2})$ from seed_{pk} and then evaluates $t'_i = \mathbf{u}_1 \cdot \mathbf{p}_{i,1} + \mathbf{u}_2 \cdot \mathbf{p}_{i,2} + \mathbf{u}_3 \cdot \mathbf{p}_{i,3}$. Therefore, verification speed largely depends on how efficiently the expanded coefficients are generated and consumed inside this loop. NoRS helps most at this point, because it simplifies public-key expansion, enabling a fusion of coefficient generation and accumulation.

We focus on the core computation $\mathbf{u} \cdot \mathbf{p}^{(i)}$, where $\mathbf{p}^{(i)}$ is a coefficient vector generated from $(\text{seed}_{\text{pk}}, i)$. In verification, this computation is used for $\mathbf{p}_{i,1}$ and $\mathbf{p}_{i,2}$, while the contribution of $\mathbf{p}_{i,3}$ is handled separately because it is already stored in the compressed public key. Algorithm 2 computes this quantity without materializing $\mathbf{p}^{(i)}$ in memory. Instead, it accumulates the dot product while the coefficients are being generated.

The kernel works block by block. For each block index j , it generates one AES output block from $(\text{seed}_{\text{pk}}, i \parallel j)$, converts it into 16 coefficients, multiplies them by the corresponding 16-entry segment of the precomputed vector $\mathbf{u}$ derived from signature, and immediately adds the result to the running sum. In this way, generation and consumption are fused into a single loop. Since $\mathbf{u}$ is padded in advance to a multiple of 16, the kernel needs no separate remainder path. This is possible because of removal of rejection sampling. In standard QR-UOV, coefficients derived from the AES output cannot be used immediately, because each occurrence of the out-of-range value q in the required prefix must be replaced

Algorithm 2 Fused expansion-and-dot-product kernel for NoRS QR-UOV verification

Input: coefficient vector $\mathbf{u} \in \mathbb{F}_q^{t_{\text{pad}}}$, public seed seed_{pk} , index $i \in [m]$

Output: $\mathbf{u} \cdot \mathbf{p}^{(i)}$, where $\mathbf{p}^{(i)}$ is the coefficient vector derived from $(\text{seed}_{\text{pk}}, i)$

```
1:  $\text{acc} \leftarrow 0$ 
2: for  $j \leftarrow 0$  to  $t_{\text{pad}}/16 - 1$  do
3:    $\mathbf{r} \leftarrow \text{AES128}(\text{seed}_{\text{pk}}, i \parallel j)$   $\triangleright \mathbf{r} \in \{0, \dots, 255\}^{16}$ 
4:    $\mathbf{p} \leftarrow \mathbf{r} \& q$   $\triangleright \mathbf{p} \in \mathbb{F}_q^{16}$ 
5:    $\text{acc} \leftarrow \text{acc} + \mathbf{u}[16j : 16j + 15] \cdot \mathbf{p}$ 
6: end for
7: return  $\text{acc}$ 
```

Note. Here, $\mathbf{p} \leftarrow \mathbf{r} \& q$ denotes byte-wise masking, and the byte value q is interpreted as $0 \in \mathbb{F}_q$.

by a later valid value. As discussed in Section 2.3, this replacement step introduces extra scanning and irregular control flow. In NoRS, by contrast, the exceptional value is mapped to 0 and can be kept as a non-canonical representative of zero in the implementation. Since $q = 2^k - 1$ with $k \in \{3, 5, 7\}$, each output yields one coefficient candidate, so each AES block yields 16 candidates. We then obtain the field coefficients by applying a fixed byte-wise mask. This regular structure allows expansion and accumulation to be merged into a single kernel.

The structure is well-suited to SIMD execution on modern x86 processors. Our implementation uses AES-NI for block generation and AVX2 for vectorized dot-product computation. Except for loading the precomputed vector $\mathbf{u}$, the main part of the kernel stays in registers, so the generated coefficients are consumed without being written back to memory. This removes the temporary buffer needed in the rejection-sampling-based implementation and reduces memory accesses during verification. A remaining practical bottleneck is the latency of AES-NI instructions. To mitigate this cost, we unroll the loop and interleave multiple independent AES blocks, increasing instruction-level parallelism and hiding part of the AES latency.

6 Performance analysis

6.1 Experimental setup

We benchmarked implementations on an Intel Skylake platform with AVX2 and AES-NI support. The processor was a Xeon Platinum 8124M running at a frequency of 3.0 GHz on an Amazon EC2 c5n.metal instance. Hyper-Threading was disabled for the benchmarked core, and the benchmark thread was pinned to a fixed core to reduce measurement noise. Cycle counts were measured with the libcpucycles [14] library using hardware performance counters. Each implementation was executed 1,000 times, and we report the median cycle count in Mcycles. The operating system was Ubuntu 24.04 LTS.

Our implementation was derived from the AVX2-optimized QR-UOV implementation in the previous paper [4]. We compared it with the pkc+skc variant of UOV¹ [5] and MAYO² [7], which we use as representative compressed-key multivariate reference schemes. Since UOV and MAYO used AES-128 in their specifications, we used the same setting in the main benchmarks to ensure a fair cross-scheme comparison. Although the original QR-UOV specification used AES-192 and AES-256 at security levels III and V, respectively, we used AES-128 for public-key expansion in these benchmarks. We also evaluated the performance with 4-round AES-128 (AES4) as supplementary measurements³. This work focused on performance optimization and did not aim to provide constant-time guarantees.

¹<https://github.com/pqov/pqov> commit 49dc53c0, avx2 implementation

²<https://github.com/PQCMayo/MAYO-C> commit 4b7cd94c, avx2 implementation

³This setting is not part of the QR-UOV or MAYO specifications, whereas it is an optional setting for UOV.

Table 5: Benchmark results of QR-UOV, UOV, and MAYO in the AES-128 setting (Mcycles)

SL	Scheme	KeyGen	Sign	Verify
I	QR-UOV AES-128	2.85	0.83	0.43
	QR-UOV AES-128 NoRS	2.73	0.72	0.30
	UOV Ip-pkc+skc	3.02	1.99	0.27
	MAYO1	0.24	0.70	0.29
III	QR-UOV AES-128	10.12	2.31	1.22
	QR-UOV AES-128 NoRS	9.79	1.98	0.87
	UOV III-pkc+skc	16.92	10.97	1.14
	MAYO3	0.56	1.45	0.65
V	QR-UOV AES-128	28.00	5.55	2.91
	QR-UOV AES-128 NoRS	27.22	4.78	2.08
	UOV V-pkc+skc	54.95	24.93	2.51
	MAYO5	1.36	3.51	1.50

For fair cross-scheme comparison, QR-UOV also uses AES-128 for public-key expansion at security levels III and V.

6.2 Benchmark results

Table 5 reports the main benchmark results under the AES-128 setting. The proposed NoRS variant consistently improves the performance of QR-UOV at all security levels. The largest gain appears in signature verification, where the speed-up is about $1.4\times$ across all security levels.

The improvement in signing is smaller but still consistent, at about $1.15\times$ across all security levels. In contrast, the effect on key generation is limited, with only a small reduction at each security level.

A comparison with other compressed-key multivariate schemes under the AES-128 setting provides an additional reference point. In verification, the cost of NoRS QR-UOV becomes close to those of UOV and MAYO at security level I. At security levels III and V, it is lower than that of UOV and higher than that of MAYO in this benchmark setting.

Table 6 gives supplementary results in the reduced-round AES-128 setting. They show the same qualitative trend more clearly: NoRS consistently improves QR-UOV at all security levels, and the largest gain appears in verification.

6.3 Profiling of verification routine

To explain the observed verification speedup, we profiled the verification routine at security level I by separating its cost into AES-based pseudorandom generation, the kernel, and the remaining operations. Table 7 reports the results under the same environment as in Section 6.1. Here, NoRS refers to the fused implementation described in Section 5, while the materialized variant is included only to separate the effect of fusion from that of removing rejection sampling. The total times in Table 7 differ slightly from the corresponding Verify times in Table 5 and Table 6, because they were measured separately with a slightly different profiling methodology.

The dominant cost lies in the AES-based pseudorandom generation and kernel parts, whereas the remaining operations stay almost constant across all variants. In the AES-128 baseline, these two parts account for about 91% of the total verification time. This confirms that QR-UOV verification is dominated by public-key expansion and the closely related kernel computation.

The main gain of NoRS already appears before fusion. Moving from the baseline to NoRS (materialized) reduces the total verification time by about 24% in the AES-128 setting and 35% in the AES4 setting, while leaving the AES cost almost unchanged. At the same time, the Kernel cost drops by about 73% in both settings, showing that the primary effect of NoRS is to eliminate the rejection-sampling

Table 6: Supplementary benchmark results in the reduced-round AES-128 setting (Mcycles)

SL	Scheme	KeyGen	Sign	Verify
I	QR-UOV AES4	2.70	0.69	0.28
	QR-UOV AES4 NoRS	2.59	0.58	0.15
	UOV Ip-pkc+skc AES4	2.90	1.88	0.15
	MAYO1 AES4	0.19	0.65	0.24
III	QR-UOV AES4	9.68	1.87	0.79
	QR-UOV AES4 NoRS	9.36	1.54	0.42
	UOV III-pkc+skc AES4	16.48	10.47	0.61
	MAYO3 AES4	0.42	1.31	0.51
V	QR-UOV AES4	26.93	4.48	1.84
	QR-UOV AES4 NoRS	26.17	3.74	0.96
	UOV V-pkc+skc AES4	53.81	23.69	1.29
	MAYO5 AES4	1.04	3.19	1.18

Table 7: Profiling breakdown of QR-UOV verification at security level I (Kcycles)

Variant	AES	Kernel	Others	Total
Baseline AES-128	248.9	127.1	37.8	413.7
NoRS AES-128 (materialized)	241.9	34.1	37.4	313.4
NoRS AES-128	252.2		34.1	286.3
Baseline AES4	103.6	125.4	38.0	266.9
NoRS AES4 (materialized)	101.9	33.7	37.9	173.4
NoRS AES4	115.0		34.0	148.9

The Kernel column includes rejection sampling and dot-product accumulation in the baseline, but only dot-product accumulation in the materialized NoRS variants. For the NoRS variants without materialization, expansion and dot-product accumulation are fused and thus reported jointly.

overhead. The fused kernel provides a further reduction of about 9% in the AES-128 setting and 14% in the AES4 setting by avoiding coefficient materialization. Overall, removing rejection sampling provides the larger gain, and fusion gives an additional improvement.

7 Discussion and conclusion

Overall, the results of this paper show that NoRS is a practical trade-off for QR-UOV. By confining the removal of rejection sampling to public-key expansion, NoRS introduces a slight bias in the generated coefficients, but for the proposed parameter sets, our analyses indicate that the claimed security levels are preserved. At the same time, the implementation becomes consistently faster, with the largest improvement in signature verification.

From the security viewpoint, the main concern is the coefficient bias introduced by deterministically mapping the unique rejected value to 0. Our analyses suggest that, for the proposed parameter sets, this bias does not meaningfully change the difficulty of the considered attacks, either asymptotically or concretely. In particular, the claw-finding, direct, reconciliation, and rectangular MinRank attacks exhibit essentially the same behavior as in the original QR-UOV. These results suggest that the induced bias is practically acceptable for the proposed parameter sets.

Table 8: Total bytes of public-key coefficients obtained by expansion

Scheme	SL = I	SL = III	SL = V
QR-UOV (127, 3)	374 868	1 147 068	2 779 245
UOV	234 872	1 036 224	2 422 464
MAYO	144 495	376 164	841 847

For UOV at security level I, we use the uov-Ip parameter set.

From the implementation viewpoint, the practical significance of NoRS lies mainly in verification. The benchmark results show that the benefit is consistent across all security levels and is much larger for verification than for key generation or signing. As a result, NoRS reduces a major source of QR-UOV-specific verification overhead and makes QR-UOV more competitive in practice among compressed-key multivariate schemes. This comparison is particularly meaningful because, unlike UOV and MAYO, QR-UOV incurs rejection sampling in public-key expansion, so the observed improvement directly reflects the reduction of this QR-UOV-specific overhead.

The particularly large speedup in verification can be explained by the structure of QR-UOV. Since verification is dominated by public-key expansion, simplifying coefficient generation has a particularly strong practical impact. Moreover, Table 8 shows that QR-UOV generates more data during public-key expansion than UOV and substantially more than MAYO at all security levels, which helps explain why the benefit of NoRS is particularly large in practice.

The supplementary AES4 results provide a complementary perspective. Once the cost of public-key expansion is reduced, QR-UOV becomes much more competitive with both UOV and MAYO in verification. This suggests that much of its practical overhead comes from coefficient generation during expansion rather than from the verification core itself.

Although our benchmarks are specific to x86 processors with AES-NI and AVX2, the NoRS idea itself is architecture-independent, since it simplifies coefficient generation at the algorithmic level. Architectures with efficient AES support and vectorized arithmetic may also benefit from the same simplification, but quantifying that benefit is left to future work.

More broadly, this work suggests that carefully scoped no-rejection simplification can be a useful design option in cryptographic schemes over odd-prime fields, provided that both their security impact and their implementation benefit are evaluated carefully.

References

- [1] H. Furue, Y. Ikematsu, Y. Kiyomura, and T. Takagi, “A new variant of unbalanced Oil and Vinegar using quotient ring: QR-UOV,” ASIACRYPT 2021, Part IV, ed. M. Tibouchi and H. Wang, LNCS, vol.13093, pp.187–217, Springer, Cham, Dec. 2021.
- [2] A. Kipnis, J. Patarin, and L. Goubin, “Unbalanced Oil and Vinegar signature schemes,” EUROCRYPT’99, ed. J. Stern, LNCS, vol.1592, pp.206–222, Springer, Berlin, Heidelberg, May 1999.
- [3] G. Alagic, M. Bros, P. Ciadoux, D. Cooper, Q. Dang, T. Dang, J.M. Kelsey, J. Lichtinger, C.A. Miller, D. Moody, R. Peralta, R. Perlner, A. Robinson, H. Silberg, D. Smith-Tone, N. Waller, and Y.K. Liu, “Status report on the first round of the additional digital signature schemes for the NIST post-quantum cryptography standardization process,” tech. rep., National Institute of Standards and Technology, 2024.
- [4] H. Amagasa, R. Ueno, and N. Homma, “AVX2 implementation of QR-UOV for modern x86 processors.” Cryptology ePrint Archive, Report 2025/1599, 2025.

- [5] W. Beullens, M. Chen, J. Ding, B. Gong, M.J. Kannwischer, J. Patarin, B. Peng, D. Schmidt, C. Shih, C. Tao, and B. Yang, “UOV — Unbalanced Oil and Vinegar,” tech. rep., National Institute of Standards and Technology, 2024.
- [6] L. Wang, C. Chou, J. Ding, Y. Kuan, J.A. Leegwater, M. Li, B. Tseng, P. Tseng, and C. Wang, “SNOVA,” tech. rep., National Institute of Standards and Technology, 2024.
- [7] W. Beullens, F. Campos, S. Celi, B. Hess, and M.J. Kannwischer, “MAYO,” tech. rep., National Institute of Standards and Technology, 2024.
- [8] R. Akiyama, H. Furue, Y. Ikematsu, F. Hoshino, K. Kinjo, H. Kosuge, S. Nakamura, S. Orihara, T. Takagi, and K. Yamakoshi, “QR-UOV,” tech. rep., National Institute of Standards and Technology, 2024.
- [9] J. Ding, B. Yang, C.O. Chen, M. Chen, and C. Cheng, “New differential-algebraic attacks and reparametrization of Rainbow,” ACNS 2008, LNCS, vol.5037, pp.242–257, Springer, 2008.
- [10] W. Beullens, “Improved cryptanalysis of UOV and Rainbow,” EUROCRYPT 2021, LNCS, vol.12696, pp.348–373, Springer, 2021.
- [11] H. Furue and Y. Ikematsu, “A new security analysis against MAYO and QR-UOV using rectangular MinRank attack,” IWSEC 2023, LNCS, vol.14128, pp.101–116, Springer, 2023.
- [12] A. Kipnis and A. Shamir, “Cryptanalysis of the oil and vinegar signature scheme,” CRYPTO 1998, LNCS, vol.1462, pp.257–266, Springer, 1998.
- [13] M. Bardet, M. Bros, D. Cabarcas, P. Gaborit, R.A. Perlner, D. Smith-Tone, J.P. Tillich, and J.A. Verbel, “Improvements of algebraic attacks for solving the rank decoding and MinRank problems,” ASIACRYPT 2020, LNCS, vol.12491, pp.507–536, Springer, 2020.
- [14] D.J. Bernstein, “libcpucycles.” <https://cpucycles.cr.yp.to/>, 2024.